

KUNUNURRA, Australia

WMO: 942160

Lat: 15.7814S

Lon: 128.7100E

Elev: 44

StdP: 100.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	12.7	-9.9	1.6	28.3	-7.0	2.1	27.5	8.4	23.4	7.9	24.1	0.9	130	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	12.9	40.2	22.3	39.2	22.5	38.2	22.7	27.6	32.2	27.1	31.6	26.7	31.2	3.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.3	29.0	26.1	21.6	28.6	25.7	21.1	28.3	88.1	32.1	85.9	31.7	84.1	31.4	36.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	DB	8.6	42.1	1.3	1.2	7.7	43.0	6.9	43.7	6.2	44.4	5.3	45.2	(4)
(5)			WB	5.9	30.1	1.0	2.9	5.1	32.2	4.5	33.9	4.0	35.5	3.2	37.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	30.0	29.6	29.6	28.3	25.4	22.7	22.5	23.9	28.0	30.7	31.7	31.0	(6)	
	DBStd	3.80	1.80	1.80	1.62	1.90	2.55	2.71	2.43	2.42	2.18	1.75	1.60	2.03	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	2	0	0	0	0	0	1	1	0	0	0	0	0	(9)	
	CDD10.0	6486	621	548	607	551	477	382	389	429	541	641	651	650	(10)	
	CDD18.3	3446	362	314	349	301	218	133	131	172	291	382	401	392	(11)	
	CDH23.3	43276	4458	3701	4206	3648	2451	1421	1544	2208	3796	5194	5521	5129	(12)	
(13)	CDH26.7	23253	2260	1793	2146	1983	1188	579	651	1106	2258	3138	3304	2847	(13)	
(14)	Wind	WSAvg	2.8	2.8	2.6	2.4	2.4	2.7	2.9	2.8	2.9	3.1	3.1	2.9	(14)	
(15) Precipitation	PrecAvg	779	185	210	136	28	5	1	2	0	2	20	69	129	(15)	
	PrecMax	1218	352	404	456	131	67	15	31	0	28	82	250	333	(16)	
	PrecMin	524	41	58	4	0	0	0	0	0	0	0	0	6	(17)	
	PrecStd	190	91	94	111	38	13	3	7	0	5	23	56	78	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	39.2	38.9	37.9	36.0	34.2	34.2	36.1	38.7	40.8	41.5	41.8	(19)	
		MCWB	23.8	24.5	23.5	22.4	21.0	20.7	19.2	19.9	20.5	21.1	21.6	22.9	(20)	
	2%	DB	38.0	37.8	37.3	36.9	34.8	33.0	33.0	34.7	37.7	39.7	40.2	39.9	(21)	
		MCWB	24.5	24.8	23.8	22.1	20.1	19.9	18.4	19.1	20.1	21.1	22.1	23.6	(22)	
	5%	DB	36.6	36.3	36.2	35.9	33.8	31.7	31.9	33.5	36.8	38.7	39.1	38.1	(23)	
		MCWB	25.2	25.3	23.6	21.9	19.8	18.6	17.6	18.0	19.9	21.2	22.5	24.5	(24)	
	10%	DB	35.1	34.8	35.1	34.8	32.5	30.2	30.6	32.2	35.8	37.6	37.9	36.5	(25)	
		MCWB	25.5	25.6	23.7	21.3	19.3	17.4	17.0	17.2	19.7	21.3	23.0	24.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	28.1	28.2	26.5	26.3	23.5	21.9	22.6	24.7	26.2	27.1	27.7	(27)	
		MCDB	33.0	32.9	32.8	31.0	29.3	29.3	30.4	31.0	30.0	31.6	31.5	32.2	(28)	
	2%	WB	27.3	27.3	27.2	25.6	24.6	21.8	20.8	21.5	23.8	25.4	26.3	27.1	(29)	
		MCDB	32.0	31.9	31.6	30.0	28.7	29.4	29.2	30.2	29.9	30.8	31.1	31.7	(30)	
	5%	WB	26.9	26.9	26.5	25.1	23.2	20.5	19.8	20.6	23.0	24.9	25.9	26.6	(31)	
		MCDB	31.4	31.3	30.7	29.5	29.0	28.3	27.6	29.1	30.0	30.5	30.9	31.3	(32)	
	10%	WB	26.5	26.4	26.0	24.6	21.9	19.4	18.5	19.6	22.4	24.3	25.5	26.2	(33)	
		MCDB	31.0	30.7	30.1	29.2	28.4	27.5	26.9	28.0	30.2	30.4	30.9	30.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	9.6	10.4	12.7	13.7	14.1	15.5	16.6	16.0	14.3	12.9	10.9	(35)	
		MCDBR	12.2	12.5	12.8	14.2	14.8	15.3	17.1	17.7	16.5	15.8	14.7	13.5	(36)	
		MCWBR	3.6	3.7	3.4	3.5	4.0	4.4	5.2	5.0	4.4	4.4	4.0	3.6	(37)	
	5% WB	MCDBR	10.1	10.1	10.0	10.7	11.3	14.0	14.7	15.5	14.8	13.5	12.6	11.3	(38)	
(39)		MCWBR	3.4	3.6	3.1	3.2	4.0	4.6	5.3	4.6	4.1	4.0	3.8	3.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.415	0.413	0.399	0.379	0.370	0.345	0.347	0.362	0.425	0.447	0.456	0.427	(40)	
	taud		2.466	2.473	2.508	2.505	2.468	2.497	2.471	2.406	2.278	2.255	2.283	2.415	(41)	
	Ebn at Noon		930	924	916	900	873	879	887	902	878	883	887	918	(42)	
	Edh at Noon		120	118	110	104	101	95	99	113	136	145	143	126	(43)	
(44)	RadAvg		6.25	6.20	6.05	5.97	5.38	5.08	5.38	6.15	6.79	7.07	6.97	6.44	(44)	
(45)	RadStd		0.50	0.76	0.69	0.39	0.35	0.22	0.19	0.17	0.27	0.37	0.29	0.41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-0.86	N/A	+0.68	+0.22	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page